

# SSS 1-3 Integrated Food Systems Laboratory • Practical Activity Cards

Progression: DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

**PA-491 Define the FEW System Boundary** Complete the investigation using explicit assumptions, systematic measurements, and...

**PA-492 Construct a Water-Flow Model** Define the objective, boundary, variables, constraints, and performance criteria. D...

**PA-493 Construct an Energy-Flow Model** Define the objective, boundary, variables, constraints, and performance criteria. D...

**PA-494 Complete a Mass Balance** Complete the investigation using explicit assumptions, systematic measurements, and...

**PA-495 Compare Supply and Demand Profiles** Use quantitative evidence and stated criteria. Compare alternatives or conditions, ...

**PA-496 Allocate a Constrained Resource** Define the objective and constraints, compare feasible alternatives, and select a s...

**PA-497 Trace a Cascade Effect** Complete the investigation using explicit assumptions, systematic measurements, and...

**PA-498 Evaluate a System Trade-Off** Use quantitative evidence and stated criteria. Compare alternatives or conditions, ...

**PA-499 Run a Scenario Analysis** Complete the investigation using explicit assumptions, systematic measurements, and...

**PA-500 Design an Integrated Decision Strategy** Define the objective, boundary, variables, constraints, and performance criteria. D...